

MTH203: Sequences and series

Lecture 05

We now explore some of the consequences of the Axiom of Completeness.

1. ARCHIMEDEAN PROPERTY

The following result was used in Example 6 in Lecture 4.

Theorem 1 (Archimedean property). *Let α be any real number. Then, there exists a natural number n such that $n > \alpha$.*

Proof. Suppose there is no natural number n such that $n > \alpha$. Then α is an upper bound of $\mathbb{N}$. By the Axiom of Completeness, $\mathbb{N}$ must have a supremum in $\mathbb{R}$. Let us denote this supremum by β .

We now use Lemma 7 from Lecture 4, which provides an alternative description of the notion of supremum. According to this description, given any $\epsilon > 0$, there exists a natural number n such that $n > \beta - \epsilon$. Taking $\epsilon = 1$, we see that there exists a natural number n such that $n > \beta - 1$. But then $n + 1 > \beta - 1 + 1 = \beta$. Since $n + 1 \in \mathbb{N}$, this contradicts the fact that $\beta = \sup(\mathbb{N})$. This contradiction shows that there exists some natural number that is greater than α . $\square$

This seemingly innocuous result has quite a few important consequences.

Corollary 2. *Let x be any real number such that $x > 0$. Then, there exists a natural number n such that $0 < 1/n < x$.*

Proof. By the Archimedean property, there exists a natural number n such that $n > 1/x$. Then $x > 1/n$. Also, since $n > 0$, $1/n > 0$. This proves the required result. $\square$

The above result shows that 0 is the infimum of the set $\{1, 1/2, 1/3, \dots\}$. This fact is often quite useful.

Before we prove the next important consequence of the Archimedean property, we note a small consequence of the Well-Ordering principle (see Theorem 5 in Lecture 3).

Theorem 3. *Let S be a subset of $\mathbb{Z}$.*

- (a) *If S has a lower bound in $\mathbb{R}$, then S has a smallest element.*
- (b) *If S has an upper bound in $\mathbb{R}$, then S has a largest element.*

Proof. We will prove (a). The proof of (b) is similar.

Let S be a subset of $\mathbb{Z}$ such that S has a lower bound α in $\mathbb{R}$. By the Archimedean property, there exists a natural number n such that $n > -\alpha$.

Now, consider the set $T = \{x + n | x \in S\}$. For any $x \in S$, we have $x > \alpha$ and hence $x - \alpha > 0$. Thus,

$$x + n > x + (-\alpha) = x - \alpha > 0.$$

This shows that every element of T is an element of $\mathbb{N}$. Since S is non-empty, T is a non-empty subset of $\mathbb{N}$. Applying the Well-Ordering Principle, we see that T has a smallest element, which we denote by n_0 . Then, $n_0 - n$ is an element of S .

We claim that it is the smallest element of S . Indeed, if x is any element of S , $x + n \in T$ and so $x + n \geq n_0$. Thus, $x \geq n_0 - n$. This completes the proof. $\square$

Remark 4. We could have also proved the above result using the Axiom of Completeness. Indeed, if $S \subset \mathbb{Z}$ is a set that has a lower bound in $\mathbb{R}$, then it must have a greatest lower bound α . One can easily show that in this case α is actually an element of S . (Can you prove this?) However, since the result is about the integers, it seems more appropriate to prove it using the Well-Ordering Principle which is

a more fundamental result. (It is actually equivalent to an axiom that is used to define the natural numbers.)

Theorem 5 (Density of $\mathbb{Q}$). *Let x and y be real numbers such that $x < y$. Then, there exists a rational number r such that $x < r < y$.*

Let us discuss the main idea behind the proof. We want to find integers m and n such that $x < m/n < y$. It is not necessarily true that there exists an integer m for any given value of n . Indeed, suppose we take $x = 1/4$ and $y = 1/3$. Then, if $n = 2$, there is no integer m such that $1/4 < m/2 < 1/3$. So, n has to be chosen appropriately first. The key observation is that the difference between m/n and $(m+1)/n$ is $1/n$. We list all the numbers of the form m/n where m is an integer.

$$\dots, -3/n, -2/n, -1/n, 0/n, 1/n, 2/n, 3/n, \dots$$

If we take any two entries in this (two-sided) sequence, the difference between them is $1/n$. These numbers march from the “left end” of the number line (i.e. from “minus infinity”) to the “right end” (i.e. to “plus infinity”). Each step is only $1/n$ in length. So if $1/n$ is smaller than the distance between x and y , we will not be able to jump over both x and y in one step. Thus, in this case, some m/n will end up between x and y . We now write this formally.

Proof. Using Corollary 2, we choose a natural number n such that $1/n < (y - x)$. Now, consider the subset S of $\mathbb{N}$ defined as follows:

$$S := \{p \in \mathbb{N} | p \geq ny\}$$

This set is non-empty (because of the Archimedean property). It has the lower bound ny in $\mathbb{R}$. Thus, by Theorem 3, this set has a smallest element, which we denote by p_0 . Note that $p_0 \geq ny$ since $p_0 \in S$.

By the definition of p_0 , we see that $p_0 - 1$ is not an element of S . Thus, $p_0 - 1 < ny$. We claim that $p_0 - 1 > nx$. Indeed, if $p_0 - 1 \leq nx$, then we have the inequalities

$$\begin{aligned} 1 &= p_0 - (p_0 - 1) \\ &\geq ny - (p_0 - 1) && (\text{since } p_0 \geq ny) \\ &\geq ny - nx && (\text{since } -(p_0 - 1) \geq -nx) \\ &= n(y - x). \end{aligned}$$

Since $1/n > 1$, this gives us $1 \cdot (1/n) \geq n(y - x) \cdot (1/n) = y - x$. However, n was chosen such that $1/n < y - x$, which gives us a contradiction. Thus, we conclude that $p_0 - 1 > nx$.

Let us take $m = p_0 - 1$. The inequalities $nx < m$ and $m < ny$ give us (after multiplication by the positive number $1/n$) the inequalities $x < m/n$ and $m/n < y$. This completes the proof. $\square$

The above theorem is often expressed by saying that $\mathbb{Q}$ is dense in $\mathbb{R}$.

2. NESTED INTERVALS

Intervals are certain subsets of the set of real numbers. Let α and β be real numbers such that $\alpha \leq \beta$.

(1) *Closed intervals:*

- (a) The set $\{x \in \mathbb{R} | \alpha \leq x \leq \beta\}$ is the closed interval $[\alpha, \beta]$.
- (b) The set $\{x \in \mathbb{R} | \alpha \leq x\}$ is the closed interval $[\alpha, \infty)$.
- (c) The set $\{x \in \mathbb{R} | x \leq \alpha\}$ is the closed interval $(-\infty, \alpha]$.

(2) *Open intervals:*

- (a) The set $\{x \in \mathbb{R} | \alpha < x < \beta\}$ is the open interval (α, β) . (This is equal to $\emptyset$ if $\alpha = \beta$.)
- (b) The set $\{x \in \mathbb{R} | \alpha < x\}$ is the open interval (α, ∞) .

- (c) The set $\{x \in \mathbb{R} | x < \alpha\}$ is the open interval $(-\infty, \alpha)$.
- (3) *Half-open intervals:*
 - (a) The set $\{x \in \mathbb{R} | \alpha \leq x < \beta\}$ is the interval $[\alpha, \beta)$.
 - (b) The set $\{x \in \mathbb{R} | \alpha < x \leq \beta\}$ is the interval $(\alpha, \beta]$.
- (4) *The real line:* $\mathbb{R}$ itself is an interval and may be denoted by $(-\infty, \infty)$. (It is considered as *open* as well as *closed*.)

In the above definitions, the elements α and β (wherever they occur) are said to be the left or right *end-points* of the intervals.

Our next result is regarding a sequence of *nested intervals*. Let $I_1, I_2, \dots$ be a sequence of intervals. We say that this is a sequence of *nested intervals* if $I_j \supset I_{j+1}$ for all j .

Theorem 6. *Let $I_1, I_2, \dots$ be a sequence of nested closed intervals where $I_j = [a_j, b_j]$ for some $a_j, b_j \in \mathbb{R}$ with $a_j \leq b_j$. Then*

$$\bigcap_{j=1}^{\infty} I_j \neq \emptyset.$$

Proof. We first prove that for any natural numbers m and n , we have $a_m \leq b_n$.

Suppose $m \leq n$. Then $I_n \leq I_m$. Thus, we have the inequalities $a_m \leq a_n \leq b_n \leq b_m$. Thus we have $a_m \leq b_n$ in this case. On the other hand, if we have $m \geq n$, then $I_m \leq I_n$ and so we have the inequalities $a_n \leq a_m \leq b_m \leq b_n$. Thus, we have $a_m \leq b_n$ in this case as well. Thus, the inequality $a_m \leq b_n$ holds in general for any two natural numbers m and n .

Now, let $S = \{a_m | m \in \mathbb{N}\}$. We have shown that for any natural number n , the number b_n is an upper bound for S . In particular, the set S is bounded above. By the Axiom of Completeness, S has a supremum. Let α be the supremum of S . Since α is the least upper bound of S , we must have $\alpha \leq b_n$ for every n . Thus, we have the inequalities $a_n \leq \alpha$ and $\alpha \leq b_n$ for every n . In other words, $\alpha \in [a_n, b_n] = I_n$. This proves the theorem. $\square$

3. READING SUGGESTIONS

This lecture corresponds to section 1.4 from the text ([1]).

REFERENCES

- [1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015. [3](#)